FMKT 256 · FALL 2026 · AI EXTRA CREDIT PROJECT

Blind Tape

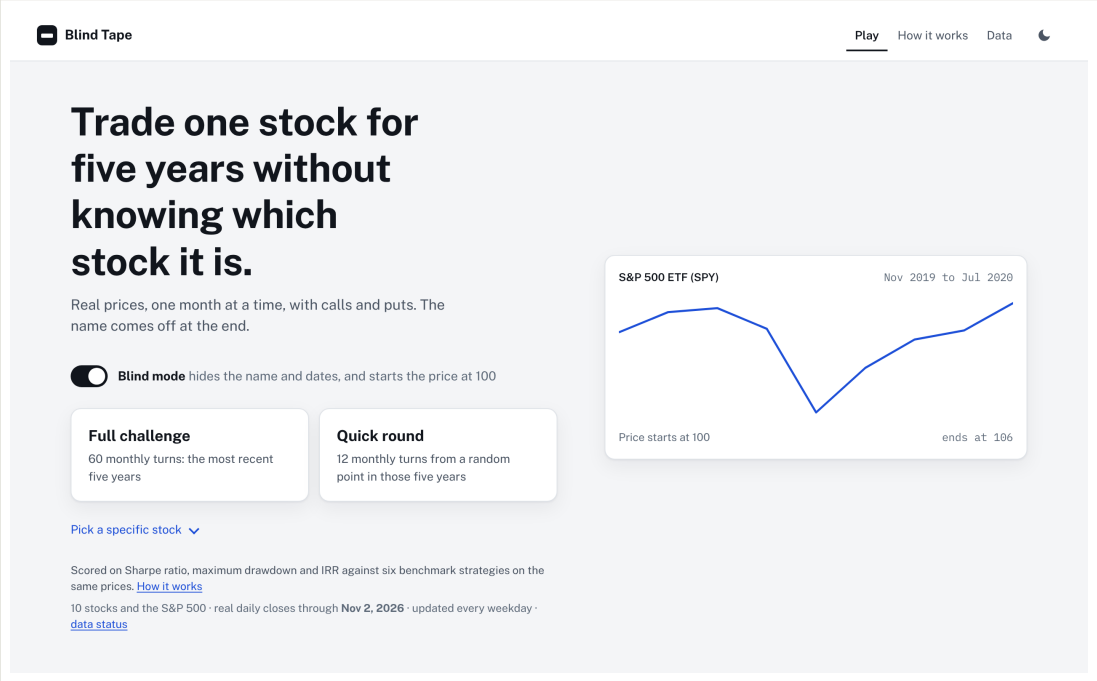
A five-year stock & options trading challenge on real market data

Prompt 1: a finance app built with AI (a trading game)

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Play the game (no login, desktop or phone)	https://fmkt256-ai-proj.vercel.app/
One-click demo game (replay)	Demo: a full 5-year blind game, scripted player
Source code, tests and data	https://github.com/cartermburke/FMKT256-AI-Proj
Daily data-update job (run history)	GitHub Actions: update-data
Live data status	Data page: series, sources, last update



What we built

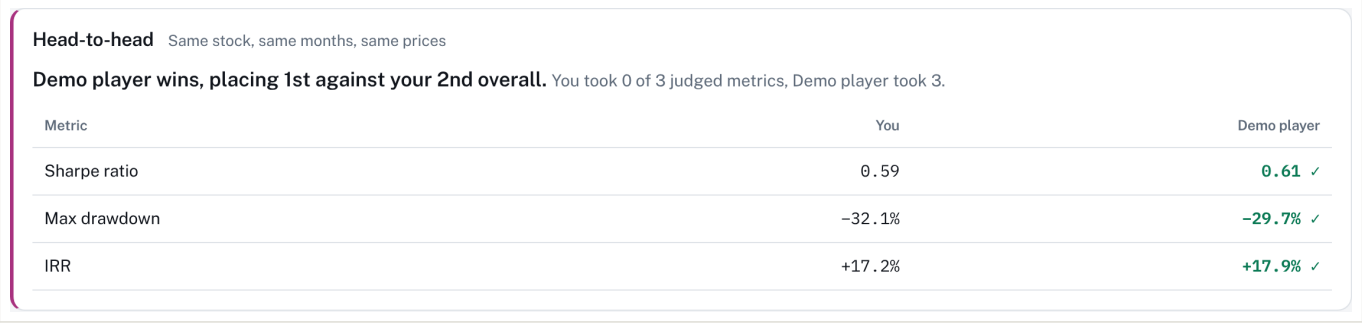
Blind Tape is a browser trading game on real market data, built with Claude. Each player starts with a fixed **$100,000** and gets a stock drawn at random from 11 instruments. In blind mode the ticker and dates are hidden and the price is rebased to 100, so no one can trade on what they remember. The game steps through the **most recent five years** of daily prices one month at a time. Each month the player sets a long or short stock position and can trade one-month **calls and puts**. At the end the stock is revealed and the player is scored on **Sharpe ratio, maximum drawdown and IRR**, the criteria named in the prompt, against six benchmark strategies played on the same data with the same costs. A friend can then be challenged to the exact same blind tape.

How it meets the prompt

- **Trading game:** fixed stake, one stock plus its derivatives over a fixed five-year period, judged on Sharpe, max drawdown and IRR, with Sortino, volatility and beta also reported.
- **Imports historical data:** about 7 years of daily adjusted closes for 11 instruments, the VIX and the 13-week T-bill yield, downloaded by code (yfinance with fallbacks) and validated before use.
- **Updates when new data arrive:** a scheduled GitHub Actions job runs every weekday after the close, appends the new bar, commits it and redeploys the site. The five-year window always ends at the latest close (currently November 2, 2026).
- **Repeatable without coding:** adding a stock to the game is one line in `config/tickers.json`; the next run downloads its history automatically.
- **No logins:** the game, the code and the data history are all public pages.

What's original

- **No hindsight.** "Trade a stock over the last five years" is easy to win if you remember NVIDIA. Blind mode hides the name and dates and rebases the price, so the game tests decisions, not memory.
- **Options without option data.** Free historical option chains don't exist, so every option is priced from that day's market: an at-the-money vol anchored to the VIX plus a volatility smile. We checked the model against the empirical evidence (delta-hedged option returns), not just intuition.
- **A score that can't be gamed.** The overall rank combines the prompt's three metrics. Drawdown enters per unit of return, so tiny positions can't win by being small. Every result sits beside six benchmark strategies on the same tape, and a rule-based coach explains what drove it.
- **Head-to-head with no server.** A challenge link carries the stock, the window and every order. A friend plays the same blind tape, and the two of you are ranked side by side.
- **Built and audited by AI agents.** An AI panel scoped all three prompts and we picked prompt 1. Thirteen adversarial rounds then tried to break the game, its redesign and this submission, and every critical and major bug got a reproduction and a regression test (page 7).



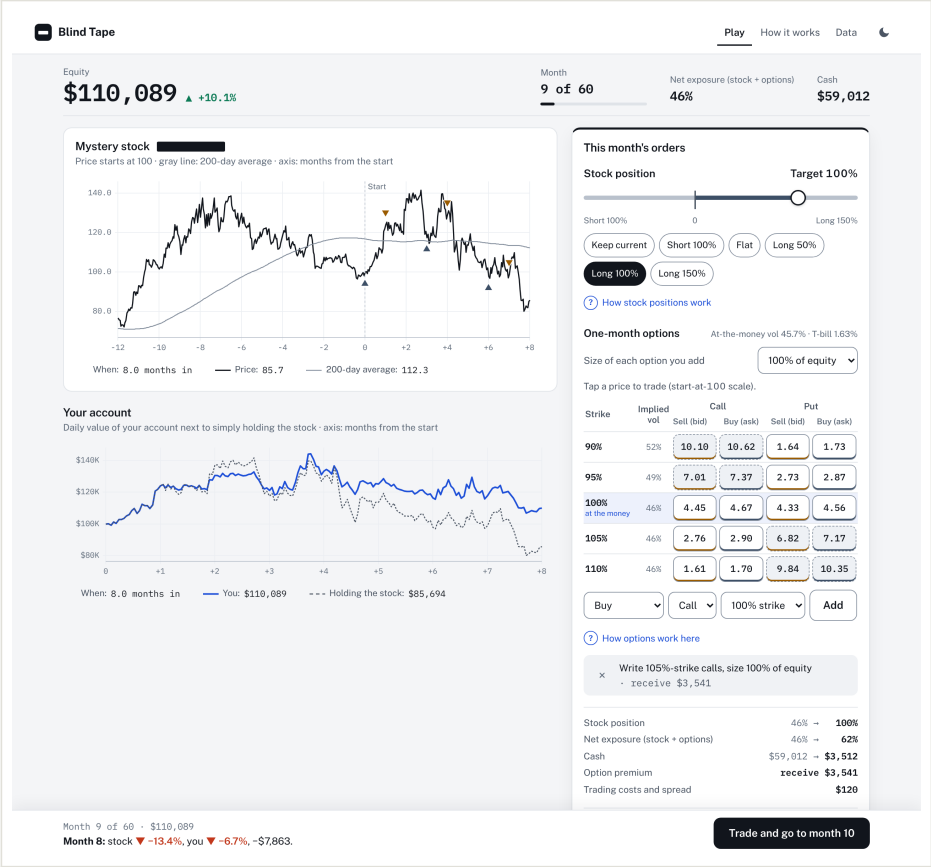
Head-to-head Same stock, same months, same prices

Demo player wins, placing 1st against your 2nd overall. You took 0 of 3 judged metrics, Demo player took 3.

Metric	You	Demo player
Sharpe ratio	0.59	0.61 ✓
Max drawdown	−32.1%	−29.7% ✓
IRR	+17.2%	+17.9% ✓

Head-to-head from a challenge link: the demo game's scripted player against a second player who let the 200-day-trend autopilot play the same tape.

Playing a month



A blind full game at month 9.

Top bar: value and return, then the month, net exposure and cash.

Price chart: from a year before the start, with the 200-day average and a marker at each trade.

Your account: its daily value, next to simply holding the stock.

Ticket: set the stock position, then tap a bid to sell or an ask to buy options at five strikes. Here the player is long 100% and has just written a 105% call against it: a covered call.

Before and after: every order is checked first, and the ticket lists each position before and after it.

Bottom bar: names the action (“Trade and go to month 10”), sums up the last month, and says why an order is blocked.

Rules

- **Turns:** 21 trading days each; 60 turns for the full challenge, or a random 12-month slice in a quick round.
- **Stock:** −100% (short) to +150% (long on margin) of equity; each trade costs 0.05% of the amount traded.
- **Options:** one-month European calls and puts at 90, 95, 100, 105 and 110% strikes, cash-settled at expiry. Up to four legs, and at most 200% of equity notional in total. You buy at the ask and sell at the bid.
- **Leverage includes options:** net exposure, meaning the stock plus every option's delta, must stay between −100% and +150% when you place an order.
- **Writing options:** written calls must be covered by shares. Written puts must be cash-secured by the cash left after the whole order.
- **Financing:** cash earns the T-bill rate daily, margin debit pays T-bill + 1.5%, shorts pay 0.5% a year to borrow, and an order can't take borrowing above 50% of equity.
- **Drift and risk controls:** limits apply to orders, and you can always hold a position the market has pushed past them. Margin is checked at every daily close, expiry days included. Equity below 25% of the gross stock position forces liquidation, and equity at zero ends the game.
- **Autopilot and replays:** an autopilot can finish the game with any benchmark rule. Every game produces a replay link that reproduces it exactly on the same data snapshot.

The scorecard and the debrief

At the end the stock is revealed. The demo game linked on page 1 is a scripted player on a blind five-year game. Its rule: long when the price is above its 200-day average and sell calls when the VIX is above 22; below the average, stay mostly in T-bills with put protection. We picked **Uber Technologies (UBER)** for the demo. The scripted player had a Sharpe ratio of **0.61** vs 0.31 for buy & hold, a max drawdown of **30%** vs 57%, and an IRR of **17.9%** vs 7.8%. Run on every stock in the pool, the same rule beat buy & hold on all three metrics for only 2 of 11 (PYPL, UBER), so the demo shows one of its best cases. That is the lesson the game teaches.

Scorecard

Place	Strategy	Sharpe	Return per drawdown	IRR	Max drawdown	Ending value	Total return	Sortino	Volatility	Beta
1	You	**0.61** #1	**0.47** #1	**+17.9%** #1	**−29.7%**	**$228,958**	**▲ +129.0%**	**0.96**	**26.7%**	**0.61**
2	200-day trend	0.59 #2	0.42 #2	+17.2% #2	−32.1%	$222,076	▲ +122.1%	0.91	26.8%	0.55
3	S&P 500	0.53 #3	0.33 #3	+12.0% #4	−24.5%	$176,452	▲ +76.5%	0.76	17.1%	1.00
4	Covered call	0.46 #4	0.27 #4	+13.6% #3	−36.6%	$190,116	▲ +90.1%	0.64	29.4%	0.88
5	Buy & hold	0.31 #5	0.07 #5	+7.8% #5	−56.7%	$145,845	▲ +45.8%	0.46	44.8%	1.33
6	T-bills	0.00 #6	0.00 #6	+3.9% #6	0.0%	$120,976	▲ +21.0%	0.00	0.1%	0.00
7	Protective put	−0.04 #7	−0.14 #7	−3.2% #7	−50.0%	$84,935	▼ −15.1%	−0.07	33.7%	0.84

The prompt judges on Sharpe, maximum drawdown and IRR. Every strategy is ranked on Sharpe, on IRR, and on return per unit of drawdown: IRR above T-bills ÷ max drawdown, a Calmar-style ratio. That is how max drawdown enters the score; raw drawdown would reward tiny positions. Strategies are ordered by the average of the three ranks (the small #1, #2 chips beside each value); equal averages go to the higher Sharpe, then the higher IRR, and only exact ties share a place. **Sharpe** is the mean daily return in excess of the T-bill rate divided by its standard deviation, × √252. **Max drawdown** is the largest peak-to-trough fall in daily equity. **IRR** is the XIRR of the account's cash flows. With a fixed stake that is the compound annual growth rate, and the engine computes a general XIRR so contributions could be added. A wiped-out account has no meaningful Sharpe and ranks last. A rule-based coach then explains the result: market timing (net of option delta), shorts, options P&L by what was bought or written, and frictions; its note on the overall placing sits beside the verdict at the top of the results.

✓ good **Market timing**

In the stock's 5 best months your net exposure (stock plus option delta) averaged 63%; in its 5 worst, 43%. Overall average: 53%.

You leaned in before rallies more than before sell-offs. That could be skill or luck, but it paid.

✓ good **Options**

You traded options in 11 of 60 months: $12,190 of premium bought and $12,862 written. Net option P&L was $6,651, or 1.3% of the starting stake a year. Bid/ask spread paid: $627.

You mostly sold options and kept more premium than you paid out. That is the volatility risk premium, and the cost is capped upside, or having the stock put to you.

• note **Frictions**

Stock trading costs $717, borrow fees $0, net interest $11,609.

Idle cash earns the 13-week T-bill rate; margin debit pays T-bill + 1.5%.

What five years of real data say about the benchmarks

Every game is benchmarked against six strategies. Here are the five risky ones on the full five-year window for every instrument in the pool (data through November 2, 2026); T-bills only always has a Sharpe of 0. Cells show the Sharpe ratio, with the best in each row in bold. The last two columns are for buy & hold.

Stock	Regime	Buy & hold	Covered call	Protective put	200-day trend	S&P 500	B&H max DD	B&H IRR
NVDA	AI moonshot	**1.02**	0.96	0.80	0.98	0.53	−66%	+54.0%
INTC	Crash and rebound	0.32	−0.05	0.37	0.50	**0.53**	−65%	+5.6%
PYPL	De-rating	−0.53	−0.41	−0.62	−0.10	**0.53**	−84%	−24.7%
MU	Memory cycle	1.10	0.73	0.93	**1.20**	0.53	−58%	+65.1%
TSLA	Wild swings	0.29	0.28	0.07	−0.44	**0.53**	−74%	+3.4%
KO	Low-vol defensive	0.71	**0.90**	0.52	0.45	0.53	−17%	+15.3%
XOM	Commodity boom	**0.89**	0.75	0.75	0.76	0.53	−21%	+27.0%
CRWD	High-vol growth	**0.66**	0.46	0.60	0.58	0.53	−68%	+28.0%
DIS	Long slide	−0.24	−0.52	−0.40	−0.67	**0.53**	−55%	−7.5%
UBER	Turnaround	0.31	0.46	−0.04	**0.59**	0.53	−57%	+7.8%
SPY	The market	0.53	**0.62**	0.32	0.30	0.53	−25%	+12.0%

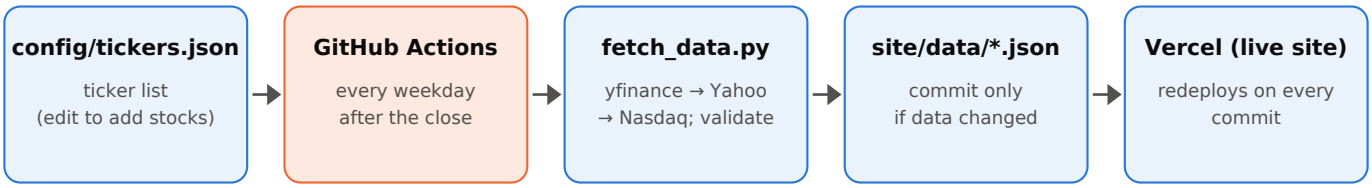
Takeaways from the same runs

- **Covered calls** beat buy & hold on Sharpe for 4 of 11 (PYPL, KO, UBER, SPY) and on total return for 5 (PYPL, TSLA, KO, UBER, SPY). They gave up the most total return on MU, NVDA, CRWD, where selling the upside cost the most.
- **Protective puts** beat buy & hold on Sharpe for 1 of 11 (INTC) and cut the maximum drawdown for 11 of 11. Insurance has a monthly cost.
- **The 200-day trend rule** beat buy & hold on Sharpe for 4 of 11 (INTC, PYPL, MU, UBER) and cut the maximum drawdown for 8 of 11. It helped most on PYPL (de-rating): Sharpe −0.10 against −0.53, and hurt most on TSLA (wild swings): Sharpe −0.44 against 0.29.
- **No single rule dominates:** 4 different strategies top the Sharpe column across the 11 instruments. So there is room for a player who reads the tape well, and that is what makes the game worth playing.

These rules trade once a month, so their results depend on the days the months start: moving the window's end by a day or two changes some of these Sharpe ratios by 0.2 or more, and the counts above shift as new data arrive.

How it works under the hood

Data pipeline: historical import and daily updates



If every source fails for a series, the last good data is kept and the site shows a stale-data banner.

Pricing options without historical option quotes

Free historical option chains for every stock don't exist, so options are priced with Black-Scholes on an implied volatility built from market data on each day:

$$IV_{atm}^{\,2} = RV_{63}(\text{stock})^2 + [\ (0.88 \cdot VIX / 100)^2 - RV_{63}(\text{S\&P 500})^2\ ]$$

$$IV(K) = IV_{atm} \cdot (1 - \text{slope} \cdot \ln(K / S))$$

RV_{63} is trailing 63-day realized volatility. The bracket is the market's variance risk premium that day: what investors pay for S&P 500 volatility beyond what the index is delivering. It is measured at the money, because the VIX is a variance-swap level that sits above at-the-money vol. The smile slope is 3.0 on both sides for the index, so its puts are richer and its calls cheaper. Single stocks get a milder 1.25 put skew and a flat call wing. **We replaced the model three times, each time because the results failed a reality check on real data.** Version 1 multiplied each stock's realized vol by the index's VIX-to-realized ratio, and covered calls beat buy & hold on almost every stock (TSLA +197% vs +32%). Version 2 priced every strike at the VIX level, and the AI review caught a 5%-out-of-the-money SPY covered call beating SPY by 46 points. Version 3 applied the index smile to single stocks, which made their out-of-the-money calls so cheap that delta-hedged call buyers profited. With the model shown here, delta-hedged buyers of out-of-the-money stock options lose on average, as the empirical evidence says, and the SPY buy-write ends within about 5 points of SPY.

Verification

- **31 unit tests** (0 failing) on the engine: Black-Scholes textbook values and put-call parity, the IV model and smile, option settlement, the leverage, covered-call and cash-secured rules, margin calls, short squeezes, busts, interest accrual, the autopilot, the ranking, and the Sharpe, Sortino, drawdown and XIRR math. They run on every push in GitHub Actions.
- **Independent cross-check:** a separate Python/pandas script recomputed 7 metrics on 66 real five-year equity curves (every benchmark on every stock): **0 mismatches** beyond 1e-6.
- **End-to-end browser test:** 165 of 165 checks pass. It plays games through the real UI in Chromium: valid and invalid orders, attempts to get around the leverage limits, autopilot after a levered month, the reveal, the replay link reproducing the exact result, damaged links, the data page, dark mode, phone and tablet screens, keyboard use and a failing network, with no console errors.
- **Adversarial review:** four separate AI agents audited the finance math, tried to break the game, played it as a first-time user, and looked for gaps. What they found and how we fixed it is on the next page.

How we used AI

- **Choosing the prompt:** Claude read the assignment and checked our real constraints: deadline, which data hosts were reachable, and existing tools. It then ran a six-agent panel. Three agents scoped one prompt each, and three judges scored the scopes as the grader, as a delivery lead and as a skeptical finance professor. The panel rated prompt 2 the safest. We chose prompt 1 because it was the most fun, and folded in the judges' warnings: test the math independently, never fake data, host on no-login pages, and fix the option-pricing bias.
- **Building:** Claude (Anthropic's Claude Code) wrote the game engine, the user interface, the data downloader and the GitHub Actions automation from a plan we reviewed and approved, and drafted this document, the README and the site's text. We chose the prompt, approved the game design, ticker pool and branding, and did the GitHub and Vercel settings, play-testing and review.
- **Testing:** Claude wrote unit tests with hand-computed answers, an independent pandas recomputation of every metric on real data, and a browser test that plays the game through the UI. Running these caught real bugs during the build. One: the 5 bp trading cost made a maximum 150% long position fail the buying-power check. Another: chart axes lost their labels.
- **Adversarial review:** after the first build, four separate AI agents audited it. One checked the finance math and looked for look-ahead bias, one tried to exploit or crash the game, and one played it as a first-time student on desktop, phone and dark mode. A fourth looked for what the first three missed. Every finding had to include a reproduction script or an exact code citation. Together they reported 35 findings, some overlapping. We reproduced each critical and major one, fixed it, wrote a regression test for it, and then reran the reviewers' own scripts against the fixed engine. A second, narrower review round on the fixed version found two flaws in our own fixes (below), and we fixed those too.
- **Redesigning the interface:** Near the end, a separate Claude session wrote a design brief for the interface, and Claude Code implemented it here: a named next-month button, a before-and-after order preview, a short reveal when a blind game ends, a one-line verdict, and colors checked for contrast in light and dark themes. The existing browser tests were the contract and every one kept passing; new checks cover the new behavior. Four independent review rounds then tried to break it (keyboard and screen reader use, phone layouts, reloads mid-game, two tabs open at once, shared challenge links). Each confirmed problem was fixed and given its own browser check before release. A five-agent audit, with a skeptic re-checking every finding, then tested the site and this document against each line of the assignment. Six more rounds re-tested each batch of fixes, the data downloader and a redesigned How it works page, until they found only small issues. We fixed the bugs they confirmed, each with its own test, and left a few small layout issues as they are.

Issues the AI review found and we fixed

- **Options got around the leverage limits.** Four 200% call legs gave about 8× delta exposure. Because Sharpe averages simple daily returns, a wiped-out account could even top the Sharpe ranking. Fix: net exposure (stock plus option delta) is capped at −100%/+150%, total option notional at 200% of equity, and a busted account has no Sharpe and ranks last.
- **Ranking by Sharpe alone misordered results.** A +20%, 0.6%-drawdown account ranked below a −76% buy & hold. Our first fix averaged the Sharpe, max-drawdown and IRR ranks, but round 2 showed it could be gamed: writing a tiny 5% put sliver ranked #1 in 39% of quick games, because raw drawdown shrinks with position size while Sharpe doesn't. Final fix: rank on Sharpe, IRR, and return per unit of drawdown (IRR above T-bills ÷ max drawdown, a Calmar-style ratio). The sliver now wins 8% of quick games, and no losing strategy ever ranks #1. Exact ties share a rank.

- **Options were mispriced.** The reviews caught two of the three pricing versions we replaced (page 6). Pricing every strike at the VIX level let a 5%-out-of-the-money SPY covered call beat SPY by 46 points. Our first fix (at-the-money index vol = 0.88 × VIX plus one linear smile) over-corrected for single stocks: round 2 showed that their out-of-the-money calls became so cheap that delta-hedged call buyers profited on all 10 stocks. Final fix: keep the steep index skew, but give single stocks a milder put skew and a flat call wing. Delta-hedged buyers of out-of-the-money stock options now lose on average, as the empirical evidence says, and the SPY buy-write ends within about 5 points of SPY.
- **Written puts weren't truly cash-secured.** Collateral was checked before the rest of the ticket, so the same cash could buy calls. Fix: the strike value must be covered by the cash left after the whole order.
- **Simply holding a 150% long was often blocked, and the autopilot crashed** (in 79% of levered starts). The borrowing cap was being re-applied every month. Fix: limits apply to orders that add risk, and drift is policed by the 25% maintenance margin, now also checked on option-expiry days. Re-running the reviewer's script gave 0 crashes in 1,298 runs.
- **Autopilot rules didn't do what their labels said** when they took over mid-game: 'T-bills only' kept your stock. Fixed and tested.
- **Data pipeline:** a run during market hours could publish an intraday price as the close, and provider rounding noise caused no-op data commits that shifted replays by a few dollars. Fixed: unfinished bars are dropped, published values are kept unless the provider truly re-adjusts them, and a backup provider only appends new days instead of rewriting the history behind shared links.
- **Usability and accessibility:** opening the autopilot on a phone broke the layout, the phone scorecard was unreadable, replay links weren't validated, a double-click could skip a month, turn results weren't announced to screen readers, and the dark-mode button contrast was too low. All fixed.
- **Smaller round-2 fixes:** a broken replay link could replace a game in progress, the coach called a median 4th of 7 'good' and told a zero-exposure player their result 'came from the stock', stale regime labels contradicted the numbers (INTC 'long decline' at +44%), and some copy didn't match the code.
- **Disclosed, not fixed:** blind mode is honor-based, because a static site has to download the price file and anyone can see it in developer tools. Option prices are model prices, not historical quotes.

Limitations

- Option prices are model prices, not historical quotes. The smile is a simple two-parameter shape, not fitted to each stock's real option market, and earnings-day volatility isn't modeled.
- Decisions are monthly and fill at the daily close. Fractional shares are allowed, and there is no market impact beyond the fixed costs.
- Prices are dividend-adjusted (total return), so options are priced as if the stock paid no dividends.
- In blind mode a well-known stock can sometimes be recognized from the shape of its chart or the era of its T-bill rate, and a player who opens the browser's developer tools can see the price file.

Run it yourself

Everything is in the public repository (https://github.com/cartermburke/FMKT256-AI-Proj). Open `site/index.html` through any static server; run `node --test tests/*.test.mjs` for the unit tests. To add a stock, edit `config/tickers.json` and push; the update job does the rest.

Academic exercise for FMKT 256, not investment advice. Results on historical data say nothing about future returns.